Double Integrals over Rectangular Regions

Name: _____ Section: _____

Before Monday: Read §5.1 and complete these notes in your own words. Show your integration steps, mark anything unclear, and bring your notes and questions to class. Attempt every prompt; identify where you get stuck.

Reading emphasis: Read the definitions, Theorems 5.1 and 5.2, and the worked iterated-integral examples carefully, especially Examples 5.2, 5.6, and 5.7. Skim the detailed Riemann-sum arithmetic in Example 5.1 and the rainfall calculation in Example 5.10. You do not need to work the end-of-section exercises.

Volumes and Double Integrals

1. For the rectangle $R = [a,b] \times [c,d]$, complete the coordinate inequalities and area formula:

$a \le x \le$ _____ $c \le y \le$ _____ $A(R) =$ _____

2. In the volume construction, explain what each factor represents:

$$f(x^*, y^*)\ \Delta A = \text{height} \times \text{base area}$$

$f(x^*, y^*)$: _____

ΔA: _____

Adding these products gives an approximation to _____.

What changes as the rectangles become smaller and more numerous?

3. Definition: double integral. In words, describe the limiting process that defines $\iint_R f(x,y)\,\mathrm{d}A$. No double-sum notation is required.

A function is **integrable over R** when this limit exists and has the same value regardless of the sample points. A function continuous on a closed rectangle is integrable there.

4. What does $\iint_R f(x,y)\,\mathrm{d}A$ represent when $f(x,y) \ge 0$ throughout R? What changes if f has both positive and negative values?

In rectangular coordinates, $\mathrm{d}A$ can be written as _____ or _____.

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Iterated Integrals

5. Definition. An iterated integral evaluates a double integral by _____.

For an inner integral with respect to y, treat _____ as constant. For an inner integral with respect to x, treat _____ as constant.

Complete these definite integrals. Each answer still contains the other variable.

$\int_0^2 (x + 2y)\,dy$ = _____ $\int_0^1 (x + 2y)\,dx$ = _____

Fubini's Theorem (Theorem 5.2)

6. If f is _____ on $R = [a, b] \times [c, d]$, then the double integral equals either iterated integral. Write both, including all four bounds and the differentials.

$$\iint_R f(x, y)\,dA = _____ = _____$$

In $dy\,dx$, integrate with respect to _____ first. In $dx\,dy$, integrate with respect to _____ first. When changing order on a rectangle, the bounds stay with their corresponding variables.

Computing in Both Orders

7. Let $f(x, y) = x + 2y$ on $R = [0, 1] \times [0, 2]$. Complete both computations. Use the worked examples in the reading as models.

(a) Integrate with respect to y first.

$$\int_0^1 \int_0^2 (x + 2y)\,dy\,dx = \int_0^1 [\qquad\qquad]\,dx = _____$$

(b) Integrate with respect to x first. Write the bounds, then evaluate.

$$\iint_R f(x, y)\,dA = _____$$

Why must the final values agree? _____

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Practice with Iterated Integrals

Show the inner antiderivative, evaluate its bounds, and then compute the outer integral. A definite inner integral should leave only the outer variable. Do not add a constant of integration.

8. Let $R = [0,2] \times [1,3]$. Sketch and label R. Write $\iint_R \left(3x - y\right) \mathrm{d}A$ in both orders, then evaluate using one order.

9. Evaluate $\int_0^2 \int_0^1 xy^2 \, \mathrm{d}y \, \mathrm{d}x$. Then write the same double integral as a product of two single integrals. Check property vi in Theorem 5.1.

10. A student writes $\int_0^2 \left(x + 2y\right) \mathrm{d}y = x^2 / 2 + 4$. Explain the error and give the correct result.

11. After reading Example 5.7, explain why one order of integration may be easier, even when both orders give the same value.

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Properties of Double Integrals (Theorem 5.1)

Assume the functions are integrable, k is constant, and $R = [a, b] \times [c, d]$. Complete each statement from the reading. Here A(R) is the area of R.

i. Linearity (sum).

$\iint_R \left(f + g \right) \mathrm{d}A =$ _____

ii. Constant multiple.

$\iint_R kf \, \mathrm{d}A =$ _____

iii. Additivity over regions.

If R = S ∪ T and their interiors do not overlap, $\iint_R f\left(x, y \right) \mathrm{d}A =$ _____

iv. Comparison.

If f ≤ g on R, then $\iint_R f\left(x, y \right) \mathrm{d}A \leq$ _____

v. Bounds.

If m ≤ f ≤ M on R, then _____ ≤ $\iint_R f\left(x, y \right) \mathrm{d}A \leq$ _____

vi. Product of functions.

If f(x,y) = g(x)h(y), then $\iint_R f\left(x, y \right) \mathrm{d}A =$ _____

Applications of Double Integrals

12. Complete the formulas. For volume, state the needed sign condition.

Area: $A\left(R \right) =$ _____ **Volume under z = f(x,y):** V = _____

The volume formula applies when _____.

Average value: $f_{\text{ave}} =$ _____

If $\iint_R f\left(x, y \right) \mathrm{d}A = 12$ and $A\left(R \right) = 4$, the average value is _____.

Questions for Monday

Write two specific questions prompted by the reading or your work. Refer to a definition, theorem, example, or step in a calculation. If the procedures seem clear, ask a "why" or "what if" question.

1. _____

2. _____